Alpha↔Omega Dynamics (AΩD)

The Hidden Temporal Dynamics of Stokes

The Art Of The Leprechaun: Fractal Calculus – $\mathfrak{S}$

43 °c

Manual Volume II — Matter I: Elements 1–118

Static relational reference

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$$A \leftrightarrow_\mu \Omega \longleftrightarrow \Omega \leftrightarrow_\mu A$$

Epitaph

$$\mathcal{N} \equiv \infty \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$$

Comprehension, cooperation, and presence are not prerequisites of AFC.

Nuff said.

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1 Manual Volume II Scope and Record Discipline

This volume is a static relational reference for **Matter I: Elements 1–118**, with eight unfilled reference slots completing the declared grouping through slot 126. It applies released AOD v41.02 [4] and the bound AFC v2.4 reference [3]. Main owns the ontology, hypotheses and mathematics; this Manual owns the elemental inputs, worked expressions, visual model, validation and historical crosswalk.

1.1 Reading route and evidence scope

The dependency order is axiom, distinction, induced relations, admitted motif, declared model, mathematical consequence. The inherited route is Main §1, Main §2, Main §3, Main §4, and Main §5. The present visual model projects declared enclosure membership and local roles. It does not instantiate the native construction trace for each element.

The active table records exact grouping and support arithmetic, assigned fixture inputs and their consequences. Its admission status is pending and its native execution status is `not_executed`. A support declaration is formal syntax; it is not a cycle or fusion certificate. Native D.E.C. rows and return witnesses remain pending as specified in §5.

1.2 Exact arithmetic and external-comparison quarantine

The support polynomial, cumulative boundaries, local counts, conditional support sums and assigned-input residuals use exact integers. Labelled-slot occupancy uses exact rationals. Arithmetic exactness, structural admission, native execution, metrology and empirical agreement are separate claims. No isotope inventory, measured calibration or empirical score is supplied here; App. B records the comparison boundary.

Provenance. The paired Main publication is built from the pinned v41.02 source before this Manual. Imported equations and sections resolve through its semantic labels. Historical v94 and v40.03r24 records are comparison inputs, with their original statuses preserved; they do not replace this foundation.

Literature note. Proof discipline and released foundation: Poyau [2, 3, 4].

2 Imported Relations and Conditional Support Arithmetic

2.1 Distinction, motif and instance evidence

Setup. Main distinguishes Null sameness from committed finite events (Main (1), Main (3)). The branch/hinge/counterbranch motif is

$$M_{\varepsilon,m} = \varepsilon\, 0_H (0_D)^m (-\varepsilon),$$

with mandatory hinge and optional dwell (Main (5)). A faithful atomic edge changes one bit of Q_4 (Main (8)). These relations provide the motif template used by the visual projection.

Derivation. For a native instance, a declaration must bind its scope, lineage and construction:

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega). \tag{1}$$

A certified closure additionally binds the retained-start return witness and at least two committed phase-bearing traversals (Main (13), Main (15)). The admitted motif record in Main (33) and traced admission in Main (37) are prerequisites, not consequences of writing an elemental label.

Specialization. This reference supplies grouping, local tuples and conditional expressions for labelled rows. It does not supply the scope-bound instance evidence in (1). Names identify the inherited distinctions or the declared reference records; they introduce no additional object or proof condition. Conversely, names cannot discharge the hypotheses of the construction they identify.

Falsification. A formatted support declaration represented as a successful certificate, or a plotted line represented as an executed edge without its trace, exceeds this profile.

Provenance. Main (5), Main (13), Main (33), Main §5.17.10.

2.2 Support polynomial and the reflection-duration domain

Setup. A support expression $\Sigma = (s_1, \ldots, s_k; L)$, with clauses $s_i = p_i{:}q_i$, is formal syntax (Main (36)). The declared finite support model is

$$\Pi(p{:}q) = q^p + q, \qquad p, q \in \mathbb{Z}_{\geq 1}. \tag{2}$$

Its provenance is Main (35) and Main (168). Admission of the proposed form remains

$$K_{\text{prop}} \xrightarrow{HT(K;B)} K_{\text{adm}}(B). \tag{3}$$

Derivation. For an admitted instance in the domain of the inherited accessor,

$$RD_\gamma(B) = \Pi^-_\gamma(B) + H_{\mu,\gamma}(B) + \Pi^+_\gamma(B). \tag{4}$$

Under equal branch supports and $H_{\mu,\gamma} = 1$,

$$RD^{(1)}_\gamma(B) = 2\Pi(p{:}q) + 1. \tag{5}$$

These are Main (31), Main (32) and Main (170). Evaluating the right-hand expression numerically does not establish its accessor prerequisites.

Specialization. Table 1 preserves the exact polynomial values. Its last column is the conditional single-hinge expression; it is not evidence that the corresponding elemental instances have been admitted.

Table 1: **Imported support polynomial and conditional single-hinge expression.**

Clause	q^p	$\Pi = q^p + q$	$2\Pi + 1$
1:2	2	4	9
3:1	1	2	5
3:2	8	10	21
3:3	27	30	61
3:4	64	68	137

The ordered count basis is

$$\mathcal{B}_{\text{curl}} = (3{:}3,\ 3{:}4,\ 1{:}2), \tag{6}$$

and its local declaration expands as

$$a{:}b{:}c :: L \equiv (3{:}3)^a :: (3{:}4)^b :: (1{:}2)^c :: L. \tag{7}$$

These local counts are distinct from the nuclear counts $(Z, A-Z)$ in §4. Main's type distinctions remain unchanged: Dimonon = 2 Monon ≠ Dion, and tetrion, tetron, tetratrion and tetrad denote different constructions.

Falsification. The single-hinge expression cannot replace (4) for unequal branches or a non-unit hinge. A declaration string cannot supply a missing construction witness or cycle-valued compression.

Provenance. Main App. B.4.1, Main App. B.1.10, Main (34), Main (40), Main (43).

3 Enclosure Membership and the Visual Model

3.1 Even enclosure sequence and cumulative boundaries

Setup. The inherited enclosure grammar distinguishes E_2 (duon), its open and seated states $E_2^0, E_2^{\pm}$, the inner E_4 tetron, and outer even support enclosures C_{2m}, $m \geq 3$. The present chart uses C_6 through C_{22} as nested enclosure interfaces. Their displayed geometry represents declared enclosure membership; no measured radius or metric scale is assigned. The semantic sources are Main §5.7 and Main App. B.3.

Derivation. The coord-0 declaration and filled boundary form are

$$0 \mid - :: Z : A : 0 : L\langle a : b : c :: L\rangle, \tag{8}$$

$$0 \mid - :: Z : A : 0 : 2m\langle m : m : 0 :: 2m\rangle. \tag{9}$$

Following Main §5.7.13,

$$\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6, \qquad \Gamma_2 = 0, \tag{10}$$

and

$$\Gamma_{m-1} < Z \leq \Gamma_m, \qquad s = Z - \Gamma_{m-1} \tag{11}$$

select the unique group and seat. Here $Z = 1, \ldots, 118$ identifies the labelled elements; integers 119–126 are unfilled reference-slot indices in the same grouping.

The local tuple is

$$a(s) = \lceil s/2 \rceil, \qquad b(s) = \lfloor s/2 \rfloor, \qquad c(s) = s \bmod 2. \tag{12}$$

Thus $c = 1$ records odd local parity and $c = 0$ even local parity. It does not determine an assigned acceptor, neutral or shedding class. For example, Li has $s = 3, c = 1$ but an assigned neutral class; O has $s = 2, c = 0$ but an assigned acceptor class. The tuple and those input classifications are distinct relations.

Specialization. The nine groups in Table 2 contain 126 reference slots, of which 118 carry element labels:

$$118/126 = 59/63, \qquad N_{\text{labelled}}(\mathsf{C}_{22})/22 = 14/22 = 7/11.$$

These are labelled-reference occupancy fractions. No retention duration or physical limit beyond 118 follows from them.

Table 2: **Declared enclosure groups and local role positions.** The last endpoint is an unfilled reference slot.

m	$2m$	Γ_{m-1}	Γ_m	Index range	Hinge $s = m$	Boundary $s = 2m$
3	6	0	6	1–6	3 Li	6 C
4	8	6	14	7–14	10 Ne	14 Si
5	10	14	24	15–24	19 K	24 Cr
6	12	24	36	25–36	30 Zn	36 Kr
7	14	36	50	37–50	43 Tc	50 Sn
8	16	50	66	51–66	58 Ce	66 Dy
9	18	66	84	67–84	75 Re	84 Po
10	20	84	104	85–104	94 Pu	104 Rf
11	22	104	126	105–126	115 Mc	126 –

3.2 Relations before layout

Setup. The visual model preserves the mapping $Z \mapsto (m, s, a, b, c)$. Within each enclosure the declared role projection assigns $1 \leq s < m$ to the first branch, $s = m$ to the hinge, $m < s < 2m$ to the counterbranch, and $s = 2m$ to the filled-tuple boundary position. This repeating organization expresses the inherited motif without supplying an executed inter-element edge.

Derivation. Nested outlines express enclosure membership; label positions express local order and role. A guide through consecutive seats is an index-order guide. A spiral may project that order while preserving membership, but its curvature is not a temporal trajectory or a transition witness. Assigned classes, when displayed in the companion or table, retain their assigned-input meaning from §4.

Falsification. Changing a preserved Z, m, s placement, confusing odd parity with an assigned acceptor state, or treating the eight unfilled slots as demonstrated physical non-retention contradicts the declared visual model. Field count $n_F \log n_F$ in Main (90) is a different scoped construction; a ring count cannot serve as its field-family witness.

Provenance. Main §5.7.10, Main §5.7.13, Main §5.8, Main §5.7.12, (11), (12).

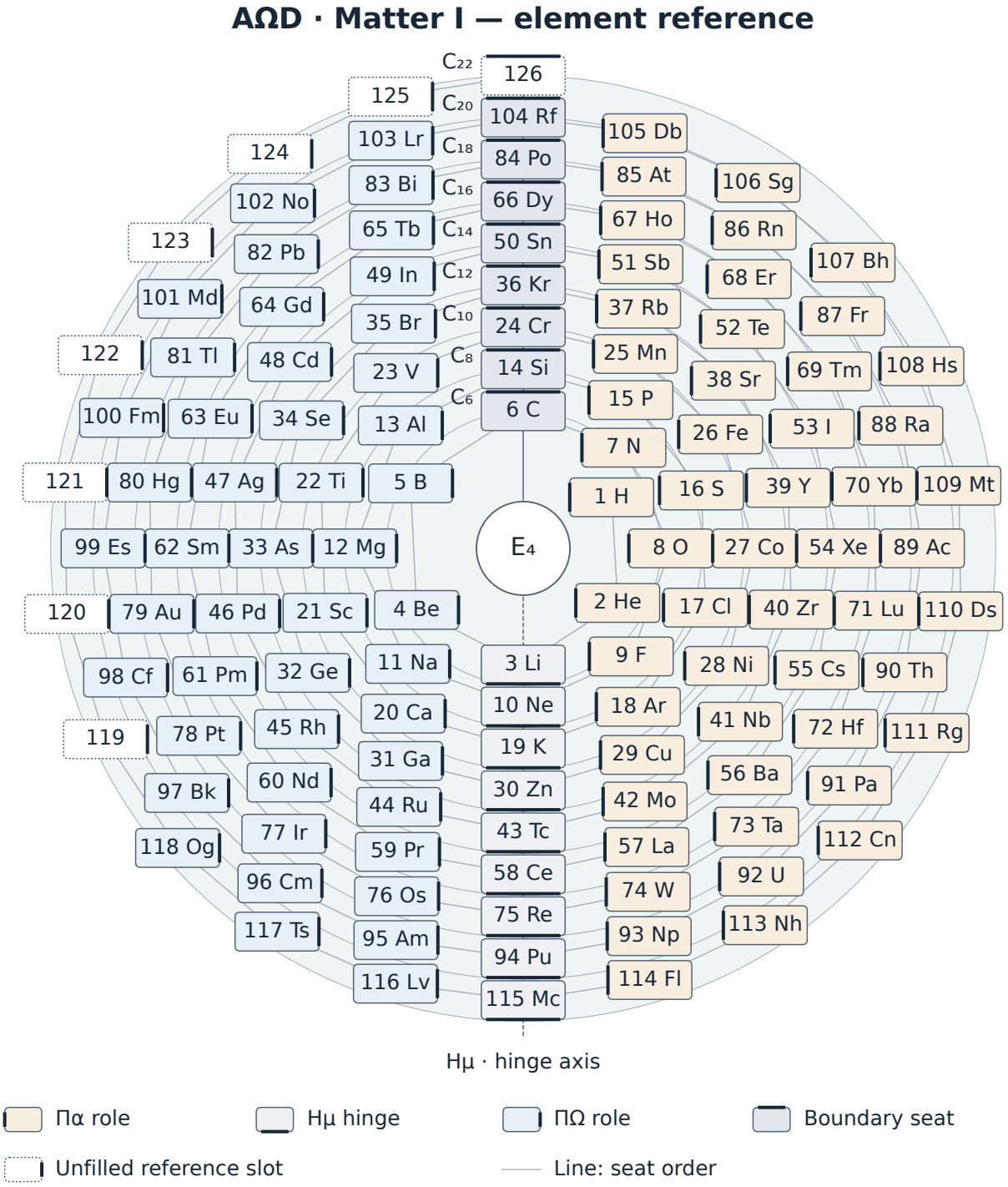


Figure 1: **Matter I static relational reference.** Nested enclosure membership and repeated branch/hinge/counterbranch roles organize all 118 element labels. The line is a seat-order guide. Positions 119–126 are unfilled reference slots in the declared grouping. Geometry supplies neither physical scale nor an executed network.

4 Conditional Support Expressions and Assigned Inputs

4.1 Inherited fusion domain and the present calculation

Setup. For admitted forms F, G with certificate-bound cycle-valued compressions $\widehat{F}, \widehat{G}$, Main defines

$$\mathrm{Fuse}_B(F, G) = \mathrm{Close}_B(\widehat{F}\,\mu\,\widehat{G}) \tag{13}$$

and

$$RD_{\mathrm{fuse}}(F, G; B) = \Pi^{-}_{\widehat{F}}(B) + H_{\mu,\widehat{F},\widehat{G}}(B) + \Pi^{+}_{\widehat{G}}(B). \tag{14}$$

Their exact predecessors are Main (40), Main (114) and Main (115). These operators retain their domains. The present elemental table supplies no inner or outer fusion certificate and does not assert an instance of (13).

Derivation. The declared reference operands use the exact polynomial values in Table 1:

$$u_{-} = 30Z + 68(A - Z), \qquad h = 1, \qquad u_{+} = 4s. \tag{15}$$

Define their conditional support sum

$$U(Z, A, s) = u_{-} + h + u_{+} = 30Z + 68(A - Z) + 1 + 4s. \tag{16}$$

This preserves the scalar expression formerly stored as `rd_fuse`. The active field `support_sum` names its established arithmetic scope. Equality to a native reflection-duration accessor additionally requires the inherited instance prerequisites.

Specialization. The nuclear count pair $(Z, A - Z)$ in (15) is distinct from the local tuple (a, b, c) in (12). At fixed Z, changing the supplied integer A leaves m, s, a, b, c unchanged and gives

$$U(Z, A + \Delta A, s) - U(Z, A, s) = 68\Delta A.$$

For carbon, $Z = s = 6$, the conditional choices $A = 12, 13, 14$ give $U = 613,681,749$. This is an algebraic illustration, not a sourced isotope catalog, stability statement or decay/fusion transition.

The reference A values are carried from the reviewed candidate's input list without alteration. Their original per-element selection policy was not recorded. The input provenance identifies that candidate source and this limitation; no broad AME or NIST citation establishes each selected A. Reference A is an integer nucleon-count input, not standard atomic weight. Projected A values for slots 119–126 remain explicitly illustrative fixture operands, not observed nuclide records. The inputs are recorded in `manual2/data/m1/reference_inputs.csv`, `model_inputs.json` and `input_provenance.json` in the same directory.

Falsification. A change in A that changes the grouping or local tuple violates this specialization. Calling a numerically correct U a certified fusion duration without its input and outer certificates exceeds its domain.

Provenance. Main (35), Main (115), (15), (12).

4.2 Assigned classifications, residuals and occupancy

Setup. The fixture keeps the candidate's assignment sets:

$$\mathcal{A} = \{1, 8, 9, 16, 17, 34, 35, 52, 53\}, \qquad \mathcal{X} = \{43, 61\} \cup \{84, \ldots, 118\}.$$

They define assigned acceptor and assigned shedding classes. They are input classifications; they are not derived from Z, A, parity, or an executed native calculation. Let $L = 2m$, $C^{\mathrm{fix}}_{\mathrm{close}} = 3L$, and $w^{\mathrm{fix}} = L$.

Derivation. With assigned pressure P^{fix}, retain the residual arithmetic

$$S^{\mathrm{fix}} = P^{\mathrm{fix}} - C^{\mathrm{fix}}_{\mathrm{close}}, \qquad X^{\mathrm{fix}} = \max(0, S^{\mathrm{fix}}). \tag{17}$$

The pressure input is $C^{\mathrm{fix}}_{\mathrm{close}} - 3$ on $\mathcal{A}$, equal to $C^{\mathrm{fix}}_{\mathrm{close}}$ on the remaining labelled neutral rows, and $C^{\mathrm{fix}}_{\mathrm{close}} + 3u_Z$ on $\mathcal{X}$, where

$$u_Z = 1 \ \ (Z = 43, 61), \qquad u_Z = \lfloor (Z - 84)/6 \rfloor + 1 \ \ (84 \le Z \le 118).$$

For illustrative unfilled slots it is $C^{\mathrm{fix}}_{\mathrm{close}} + 3(Z - 118)$. Assigned duration values are respectively L, $L+2$, $L-1$, and 0. Assigned route and Yro labels are carried as fixture data. These values do not demonstrate retention, instability, transfer or non-retention.

The arithmetic uses the form of Main (107) and Main (108); it does not supply their native pressure, field and window evidence. Likewise, a declared route label does not execute the sheddic partition, lift or higher-scope fusion in Main (110), Main (119) and Main (121).

Specialization. The 118 labelled rows have disjoint assigned residual classes: 72 neutral, 9 deficit/acceptor and 37 positive-surplus/shedding. Boundary position is a separate attribute: eight labelled endpoints are $\{6, 14, 24, 36, 50, 66, 84, 104\}$. Po and Rf are both endpoints and assigned shedding rows. Adding endpoint counts to route-class counts would double-count those attributes. Of the 72 neutral rows, six are endpoints and 66 are not.

The C_{22} fraction 7/11 records 14 labelled slots out of 22. Slots 119–126 are unfilled in this reference; their assigned fixture values establish neither a physical cutoff nor an impossibility of further elements.

Falsification. The assignment table and its equations must agree exactly. Positive X^{fix} for negative S^{fix}, or a changed assigned set without a new input binding, fails the fixture. Agreement with these definitions is not empirical validation.

Provenance. Main (107), Main (108), Main §5.7.12, Main §5.6.6.

4.3 Complete reference ledger

The authoritative generated ledger is `manual2/data/m1/matter1_elements_1_118_ledger.csv`. It binds each static declaration to its input record, distinguishes labelled elements from unfilled slots, and carries pending admission and unexecuted native status. The support declaration is syntax; the support sum and assigned fields retain the scopes above. Assigned route codes are `neutral`, `deficit`, `surplus` and `unfilled`; group-boundary and named-anchor flags are independent attributes. The generated table and figure consume the same validated data. Declared compatibility copies must compare equal.

Table 3: **Complete declared reference and conditional support ledger.** $Z = 1 \dots 118$ names supplied elements; slots $119 \dots 126$ are unfilled reference positions. Pressure and route classes are assigned inputs; the support sum is a conditional expression, not an admitted native duration. Grouping follows Main §5.7.13; the expression specializes Main (35).

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\mathrm{fix}}, X^{\mathrm{fix}}$	Assigned class
1	H	1		C_6	1 : 0 : 1	30	1	4	35	15	18	$-3, 0$	Deficit
2	He	4		C_6	1 : 1 : 0	196	1	8	205	18	18	$0, 0$	Neutral
3	Li	7		C_6	2 : 1 : 1	362	1	12	375	18	18	$0, 0$	Neutral
4	Be	9		C_6	2 : 2 : 0	460	1	16	477	18	18	$0, 0$	Neutral
5	B	11		C_6	3 : 2 : 1	558	1	20	579	18	18	$0, 0$	Neutral
6	C	12		C_6	3 : 3 : 0	588	1	24	613	18	18	$0, 0$	Neutral + anchor
7	N	14		C_8	1 : 0 : 1	686	1	4	691	24	24	$0, 0$	Neutral
8	O	16		C_8	1 : 1 : 0	784	1	8	793	21	24	$-3, 0$	Deficit
9	F	19		C_8	2 : 1 : 1	950	1	12	963	21	24	$-3, 0$	Deficit
10	Ne	20		C_8	2 : 2 : 0	980	1	16	997	24	24	$0, 0$	Neutral
11	Na	23		C_8	3 : 2 : 1	1146	1	20	1167	24	24	$0, 0$	Neutral
12	Mg	24		C_8	3 : 3 : 0	1176	1	24	1201	24	24	$0, 0$	Neutral
13	Al	27		C_8	4 : 3 : 1	1342	1	28	1371	24	24	$0, 0$	Neutral
14	Si	28		C_8	4 : 4 : 0	1372	1	32	1405	24	24	$0, 0$	Neutral + anchor
15	P	31		C_{10}	1 : 0 : 1	1538	1	4	1543	30	30	$0, 0$	Neutral
16	S	32		C_{10}	1 : 1 : 0	1568	1	8	1577	27	30	$-3, 0$	Deficit
17	Cl	35		C_{10}	2 : 1 : 1	1734	1	12	1747	27	30	$-3, 0$	Deficit
18	Ar	40		C_{10}	2 : 2 : 0	2036	1	16	2053	30	30	$0, 0$	Neutral
19	K	39		C_{10}	3 : 2 : 1	1930	1	20	1951	30	30	$0, 0$	Neutral
20	Ca	40		C_{10}	3 : 3 : 0	1960	1	24	1985	30	30	$0, 0$	Neutral
21	Sc	45		C_{10}	4 : 3 : 1	2262	1	28	2291	30	30	$0, 0$	Neutral
22	Ti	48		C_{10}	4 : 4 : 0	2428	1	32	2461	30	30	$0, 0$	Neutral
23	V	51		C_{10}	5 : 4 : 1	2594	1	36	2631	30	30	$0, 0$	Neutral
24	Cr	52		C_{10}	5 : 5 : 0	2624	1	40	2665	30	30	$0, 0$	Neutral + anchor
25	Mn	55		C_{12}	1 : 0 : 1	2790	1	4	2795	36	36	$0, 0$	Neutral
26	Fe	56		C_{12}	1 : 1 : 0	2820	1	8	2829	36	36	$0, 0$	Neutral
27	Co	59		C_{12}	2 : 1 : 1	2986	1	12	2999	36	36	$0, 0$	Neutral
28	Ni	58		C_{12}	2 : 2 : 0	2880	1	16	2897	36	36	$0, 0$	Neutral
29	Cu	63		C_{12}	3 : 2 : 1	3182	1	20	3203	36	36	$0, 0$	Neutral
30	Zn	64		C_{12}	3 : 3 : 0	3212	1	24	3237	36	36	$0, 0$	Neutral
31	Ga	69		C_{12}	4 : 3 : 1	3514	1	28	3543	36	36	$0, 0$	Neutral
32	Ge	74		C_{12}	4 : 4 : 0	3816	1	32	3849	36	36	$0, 0$	Neutral
33	As	75		C_{12}	5 : 4 : 1	3846	1	36	3883	36	36	$0, 0$	Neutral
34	Se	80		C_{12}	5 : 5 : 0	4148	1	40	4189	33	36	$-3, 0$	Deficit
35	Br	79		C_{12}	6 : 5 : 1	4042	1	44	4087	33	36	$-3, 0$	Deficit
36	Kr	84		C_{12}	6 : 6 : 0	4344	1	48	4393	36	36	$0, 0$	Neutral + anchor
37	Rb	85		C_{14}	1 : 0 : 1	4374	1	4	4379	42	42	$0, 0$	Neutral
38	Sr	88		C_{14}	1 : 1 : 0	4540	1	8	4549	42	42	$0, 0$	Neutral
39	Y	89		C_{14}	2 : 1 : 1	4570	1	12	4583	42	42	$0, 0$	Neutral
40	Zr	90		C_{14}	2 : 2 : 0	4600	1	16	4617	42	42	$0, 0$	Neutral
41	Nb	93		C_{14}	3 : 2 : 1	4766	1	20	4787	42	42	$0, 0$	Neutral
42	Mo	98		C_{14}	3 : 3 : 0	5068	1	24	5093	42	42	$0, 0$	Neutral
43	Tc	98		C_{14}	4 : 3 : 1	5030	1	28	5059	45	42	$3, 3$	Surplus
44	Ru	102		C_{14}	4 : 4 : 0	5264	1	32	5297	42	42	$0, 0$	Neutral
45	Rh	103		C_{14}	5 : 4 : 1	5294	1	36	5331	42	42	$0, 0$	Neutral
46	Pd	106		C_{14}	5 : 5 : 0	5460	1	40	5501	42	42	$0, 0$	Neutral
47	Ag	107		C_{14}	6 : 5 : 1	5490	1	44	5535	42	42	$0, 0$	Neutral
48	Cd	114		C_{14}	6 : 6 : 0	5928	1	48	5977	42	42	$0, 0$	Neutral
49	In	115		C_{14}	7 : 6 : 1	5958	1	52	6011	42	42	$0, 0$	Neutral
50	Sn	120		C_{14}	7 : 7 : 0	6260	1	56	6317	42	42	$0, 0$	Neutral + anchor
51	Sb	121		C_{16}	1 : 0 : 1	6290	1	4	6295	48	48	$0, 0$	Neutral
52	Te	130		C_{16}	1 : 1 : 0	6864	1	8	6873	45	48	$-3, 0$	Deficit
53	I	127		C_{16}	2 : 1 : 1	6622	1	12	6635	45	48	$-3, 0$	Deficit
54	Xe	132		C_{16}	2 : 2 : 0	6924	1	16	6941	48	48	$0, 0$	Neutral
55	Cs	133		C_{16}	3 : 2 : 1	6954	1	20	6975	48	48	$0, 0$	Neutral
56	Ba	138		C_{16}	3 : 3 : 0	7256	1	24	7281	48	48	$0, 0$	Neutral
57	La	139		C_{16}	4 : 3 : 1	7286	1	28	7315	48	48	$0, 0$	Neutral
58	Ce	140		C_{16}	4 : 4 : 0	7316	1	32	7349	48	48	$0, 0$	Neutral
59	Pr	141		C_{16}	5 : 4 : 1	7346	1	36	7383	48	48	$0, 0$	Neutral
60	Nd	142		C_{16}	5 : 5 : 0	7376	1	40	7417	48	48	$0, 0$	Neutral
61	Pm	145		C_{16}	6 : 5 : 1	7542	1	44	7587	51	48	$3, 3$	Surplus

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\mathrm{fix}}, X^{\mathrm{fix}}$	Assigned class
62	Sm	152		C_{16}	6 : 6 : 0	7980	1	48	8029	48	48	0, 0	Neutral
63	Eu	153		C_{16}	7 : 6 : 1	8010	1	52	8063	48	48	0, 0	Neutral
64	Gd	158		C_{16}	7 : 7 : 0	8312	1	56	8369	48	48	0, 0	Neutral
65	Tb	159		C_{16}	8 : 7 : 1	8342	1	60	8403	48	48	0, 0	Neutral
66	Dy	164		C_{16}	8 : 8 : 0	8644	1	64	8709	48	48	0, 0	Neutral + anchor
67	Ho	165		C_{18}	1 : 0 : 1	8674	1	4	8679	54	54	0, 0	Neutral
68	Er	166		C_{18}	1 : 1 : 0	8704	1	8	8713	54	54	0, 0	Neutral
69	Tm	169		C_{18}	2 : 1 : 1	8870	1	12	8883	54	54	0, 0	Neutral
70	Yb	174		C_{18}	2 : 2 : 0	9172	1	16	9189	54	54	0, 0	Neutral
71	Lu	175		C_{18}	3 : 2 : 1	9202	1	20	9223	54	54	0, 0	Neutral
72	Hf	180		C_{18}	3 : 3 : 0	9504	1	24	9529	54	54	0, 0	Neutral
73	Ta	181		C_{18}	4 : 3 : 1	9534	1	28	9563	54	54	0, 0	Neutral
74	W	184		C_{18}	4 : 4 : 0	9700	1	32	9733	54	54	0, 0	Neutral
75	Re	187		C_{18}	5 : 4 : 1	9866	1	36	9903	54	54	0, 0	Neutral
76	Os	192		C_{18}	5 : 5 : 0	10168	1	40	10209	54	54	0, 0	Neutral
77	Ir	193		C_{18}	6 : 5 : 1	10198	1	44	10243	54	54	0, 0	Neutral
78	Pt	195		C_{18}	6 : 6 : 0	10296	1	48	10345	54	54	0, 0	Neutral
79	Au	197		C_{18}	7 : 6 : 1	10394	1	52	10447	54	54	0, 0	Neutral
80	Hg	202		C_{18}	7 : 7 : 0	10696	1	56	10753	54	54	0, 0	Neutral
81	Tl	205		C_{18}	8 : 7 : 1	10862	1	60	10923	54	54	0, 0	Neutral
82	Pb	208		C_{18}	8 : 8 : 0	11028	1	64	11093	54	54	0, 0	Neutral
83	Bi	209		C_{18}	9 : 8 : 1	11058	1	68	11127	54	54	0, 0	Neutral
84	Po	209		C_{18}	9 : 9 : 0	11020	1	72	11093	57	54	3, 3	Surplus + anchor
85	At	210		C_{20}	1 : 0 : 1	11050	1	4	11055	63	60	3, 3	Surplus
86	Rn	222		C_{20}	1 : 1 : 0	11828	1	8	11837	63	60	3, 3	Surplus
87	Fr	223		C_{20}	2 : 1 : 1	11858	1	12	11871	63	60	3, 3	Surplus
88	Ra	226		C_{20}	2 : 2 : 0	12024	1	16	12041	63	60	3, 3	Surplus
89	Ac	227		C_{20}	3 : 2 : 1	12054	1	20	12075	63	60	3, 3	Surplus
90	Th	232		C_{20}	3 : 3 : 0	12356	1	24	12381	66	60	6, 6	Surplus
91	Pa	231		C_{20}	4 : 3 : 1	12250	1	28	12279	66	60	6, 6	Surplus
92	U	238		C_{20}	4 : 4 : 0	12688	1	32	12721	66	60	6, 6	Surplus
93	Np	237		C_{20}	5 : 4 : 1	12582	1	36	12619	66	60	6, 6	Surplus
94	Pu	244		C_{20}	5 : 5 : 0	13020	1	40	13061	66	60	6, 6	Surplus
95	Am	243		C_{20}	6 : 5 : 1	12914	1	44	12959	66	60	6, 6	Surplus
96	Cm	247		C_{20}	6 : 6 : 0	13148	1	48	13197	69	60	9, 9	Surplus
97	Bk	247		C_{20}	7 : 6 : 1	13110	1	52	13163	69	60	9, 9	Surplus
98	Cf	251		C_{20}	7 : 7 : 0	13344	1	56	13401	69	60	9, 9	Surplus
99	Es	252		C_{20}	8 : 7 : 1	13374	1	60	13435	69	60	9, 9	Surplus
100	Fm	257		C_{20}	8 : 8 : 0	13676	1	64	13741	69	60	9, 9	Surplus
101	Md	258		C_{20}	9 : 8 : 1	13706	1	68	13775	69	60	9, 9	Surplus
102	No	259		C_{20}	9 : 9 : 0	13736	1	72	13809	72	60	12, 12	Surplus
103	Lr	266		C_{20}	10 : 9 : 1	14174	1	76	14251	72	60	12, 12	Surplus
104	Rf	267		C_{20}	10 : 10 : 0	14204	1	80	14285	72	60	12, 12	Surplus + anchor
105	Db	268		C_{22}	1 : 0 : 1	14234	1	4	14239	78	66	12, 12	Surplus
106	Sg	269		C_{22}	1 : 1 : 0	14264	1	8	14273	78	66	12, 12	Surplus
107	Bh	270		C_{22}	2 : 1 : 1	14294	1	12	14307	78	66	12, 12	Surplus
108	Hs	277		C_{22}	2 : 2 : 0	14732	1	16	14749	81	66	15, 15	Surplus
109	Mt	278		C_{22}	3 : 2 : 1	14762	1	20	14783	81	66	15, 15	Surplus
110	Ds	281		C_{22}	3 : 3 : 0	14928	1	24	14953	81	66	15, 15	Surplus
111	Rg	282		C_{22}	4 : 3 : 1	14958	1	28	14987	81	66	15, 15	Surplus
112	Cn	285		C_{22}	4 : 4 : 0	15124	1	32	15157	81	66	15, 15	Surplus
113	Nh	286		C_{22}	5 : 4 : 1	15154	1	36	15191	81	66	15, 15	Surplus
114	Fl	289		C_{22}	5 : 5 : 0	15320	1	40	15361	84	66	18, 18	Surplus
115	Mc	290		C_{22}	6 : 5 : 1	15350	1	44	15395	84	66	18, 18	Surplus
116	Lv	293		C_{22}	6 : 6 : 0	15516	1	48	15565	84	66	18, 18	Surplus
117	Ts	294		C_{22}	7 : 6 : 1	15546	1	52	15599	84	66	18, 18	Surplus
118	Og	294		C_{22}	7 : 7 : 0	15508	1	56	15565	84	66	18, 18	Surplus
119	–	297		C_{22}	8 : 7 : 1	15674	1	60	15735	69	66	3, 3	Unfilled fixture
120	–	298		C_{22}	8 : 8 : 0	15704	1	64	15769	72	66	6, 6	Unfilled fixture
121	–	301		C_{22}	9 : 8 : 1	15870	1	68	15939	75	66	9, 9	Unfilled fixture
122	–	302		C_{22}	9 : 9 : 0	15900	1	72	15973	78	66	12, 12	Unfilled fixture
123	–	305		C_{22}	10 : 9 : 1	16066	1	76	16143	81	66	15, 15	Unfilled fixture
124	–	306		C_{22}	10 : 10 : 0	16096	1	80	16177	84	66	18, 18	Unfilled fixture
125	–	309		C_{22}	11 : 10 : 1	16262	1	84	16347	87	66	21, 21	Unfilled fixture
126	–	310		C_{22}	11 : 11 : 0	16292	1	88	16381	90	66	24, 24	Unfilled fixture

5 Verification Scope and Pending Native Evidence

5.1 Executed static checks and inherited audit obligations

Setup. The static checker verifies stored inputs and generated artifacts without rewriting them. Generation is a separate explicit operation. Exact grouping, local tuples, support sums, assignment residuals, identities and data/figure agreement are testable here. Their success does not certify the six native hooks in Main §5.17.

Derivation. Table 4 separates the available static checks from the instance evidence required for native claims. A check result is reported only after that check executes; this table describes scope, not an unrecorded run count.

Table 4: **Verification scope.** Native obligations remain pending in this release.

Inherited hook	Static reference check	Native evidence not supplied
Spine	Declared role mapping and operand sum	Retained occurrences at each finite depth
Seat/enclosure	Nine boundaries, tuples and labelled occupancy	Scoped field-family and routing witness
Sheddic path	Assigned class, route and residual consistency	Realized path and retention window
Cycle	Support syntax and pending status	Construction, retained start and return certificate
Fusion	Conditional operand and residual equations	Admitted inner forms and outer closure certificate
Naming	Stable declarations, record kinds and input bindings	No name substitutes for instance evidence

Specialization. A future native D.E.C. record has the inherited form

$$(B,\ v,\ e,\ v_e,\ \sigma(e),\ \mathrm{adm}(e;B),\ w(e;B),\ P(e;B),\ \mathrm{route}(e;B)).$$

The present reference ledger does not instantiate it. Native execution would require bound Q_4 Hamming-1 traversals, scope and window, admissibility and exact weights, normalized kernels on nonempty positive-weight support, and reversed-edge SADAR pairs:

$$\sum_e P(e;B) = 1, \qquad \mathrm{adm}(e;B) = 0 \Rightarrow P(e;B) = 0, \qquad \mathfrak{S}_B(C_{\bar{e}}) = -\mathfrak{S}_B(C_e).$$

An empty admissible support does not define a normalized kernel. Retained-return and fusion witnesses must bind the same identities, scopes and lineages as their consumers. None of these obligations is discharged by this reference's assignment tables.

Falsification. Actual negative probes must reject changed values, duplicate or missing identities, mismatched mirrors, invalid status promotion and altered provenance without replacing the input bytes. In particular, a forged certificate-like string must not turn pending admission into a pass. Unknown command flags must fail, and the verification command must not regenerate corrupted input.

Publication checks bind Main source/PDF/AUX/FLS before resolving semantic references. Figure checks cover all 126 marks at final size, including the eight unfilled slots. These are distinct from native dynamics or empirical tests.

Provenance. Main §5.17.1, Main §5.17.4, Main §5.17.5, Main §5.17.7, Main §5.17.8, Main §5.17.10, Main (66), Main (95).

Literature note. D.E.C. row grammar and execution protocol: released AOD Manual v41.02, §4 [4].

6 Reference, History and Pending-Comparison Registry

Setup. This registry distinguishes active static calculations, preserved historical declarations and pending native or measured comparisons. It does not classify every integer table as a completed native simulation.

Table 5: **Matter I evidence inventory.**

Record	Present scope	Completion boundary
Support polynomial	Exact integer expression	Does not admit a motif
Enclosure groups and local tuples	Exact grouping and role projection	Does not execute a path
Reference-A and assigned classes	Pinned candidate fixture inputs	Original selection provenance unresolved
Conditional support and residuals	Exact consequences of those inputs	Native accessor/retention claims pending
v94 placement crosswalk	118 rows, six compared fields	708 comparisons; separate from lane inventory
v94 observable inventory	118 elements across E0–E5	Preserve original per-lane statuses
r24 occurrence/packet crosswalk	Historical declaration identities	Native transition and SADAR execution deferred
Native D.E.C. and certificates	Required schema and obligations	Not executed; admission pending
Isotope, mass and spectral comparisons	Future sourced input/comparison lanes	No catalog, calibration or scores supplied

Specialization. The original v94 observable matrix has 708 rows: 118 E0 identity/registry rows marked `active_registry`, and 118 rows in each of E1 placement/classification, E2 atomic mass/standard atomic weight, E3 isotope support, E4 spectra and E5 reaction/routes marked `declared_lane_not_scored_in_v94`. Its ten value maps remain `deferred_not_attached`. These statuses are separate from the successful six-field placement comparison.

The r24 crosswalk retains occurrence, transition-packet and SADAR declaration IDs as historical IDs. Its execution remains declared but unexecuted; target values are not read, measured reports are closed and scores absent. It does not contain source-to-destination event witnesses. Correspondence of the original and public atlas is bound by parsed manifest content and five matching file hashes; it is not an identity claim about an entire public repository.

Falsification. A historical pending lane promoted to a current execution result without new evidence, or a hash comparison described as scientific admission, violates the inventory. Preserve predecessor status and current status in distinct fields. Prerecorded counterfactual outcomes do not count as actual hostile tests.

Provenance. The release provenance records the reviewed PR, original v94/r24 archive hashes, extracted member hashes, the six-field mapping, historical identifier joins and the current input bindings. Small verified source extracts and the generated `declaration_crosswalk.csv`, `observable_coverage.csv` and `value_map_status.csv` are stored under `manual2/data/m1/history/`. Their bindings are in `manual2/evidence/historical_sources.json`; current input hashes are in `manual2/evidence/input_manifest.json`. Exact static checks, source identity checks and any future scientific claims retain separate result scopes.

A Exact Arithmetic Fixtures

A.1 Scope

These are arithmetic fixtures. They do not claim simulated native trajectories or certified closure. The inherited four-step convention is in Main App. I; polynomial support is in Main App. H.

A.2 Four-step decompositions

For $L = 4c_L + r_L$, $r_L \in \{0, 1, 2, 3\}$, the outer even indices give

$$\begin{aligned} 6 &= 4(1) + 2, & 8 &= 4(2) + 0, & 10 &= 4(2) + 2, \\ 12 &= 4(3) + 0, & 14 &= 4(3) + 2, & 16 &= 4(4) + 0, \\ 18 &= 4(4) + 2, & 20 &= 4(5) + 0, & 22 &= 4(5) + 2. \end{aligned}$$

Even L has remainder 0 or 2; an odd index has remainder 1 or 3. This division identity is not a theorem that every closure has four steps. The inherited history-weighted load requires its declared histories and weights (Main (172)).

A.3 Labelled-reference occupancy

The 126 declared slots and 118 element labels give

$$\frac{118}{126} = \frac{59}{63}, \qquad \frac{N_{\mathrm{labelled}}(\mathsf{C}_{22})}{22} = \frac{14}{22} = \frac{7}{11}, \qquad \frac{N_{\mathrm{unfilled}}(\mathsf{C}_{22})}{22} = \frac{8}{22} = \frac{4}{11}.$$

These are exact occupancy fractions, independent of assigned TTL values and native retention.

A.4 Falsification

Non-integer quotient/remainder, an invalid remainder, or disagreement with the labelled-slot counts fails the corresponding arithmetic fixture. It does not by itself falsify a native execution that has not been performed.

Provenance. Main App. I.2, Main App. I.4, (10).

B Pending External Comparison

B.1 Scope and numeric policy

No measured comparison is executed in this release. A future comparison must bind an established candidate, its declared value map and a sourced target record before evaluating

$$\mathsf{Q}_{\mathrm{ext}}(\nu_{Z,A\leftrightarrow\Omega};\mathcal{T}_Z). \tag{18}$$

A conditional support sum does not become a measured prediction merely by being frozen. Candidate selection and the mathematical/physical meaning of the value map must be explicit before joining targets.

External mass, spectral or isotope data retain their sources and units. A displayed measured decimal requires

$$(x_{\mathrm{disp}}, u_x, \mathrm{unit}(x), \mathrm{src}(x), \mathrm{round}(x), \mathrm{scope}(x)). \tag{19}$$

These records are distinct from the exact integer reference A and from $(Z, A-Z)$. AME and NIST references below identify possible comparison sources; no per-element selection of the present A inputs is attributed to them.

B.2 Pending comparison gates

A future measured lane must identify the candidate and frozen inputs, applicable admission/execution evidence, value map, target version and row identity, units and precision, comparison formula, and resulting status. Missing inputs remain explicitly pending. No spectral, isotope-stability, charge, decay or reaction agreement is inferred from the reference figure.

Provenance. Main §5.16.12; released AOD Manual v41.02, §1.7–§1.8 and App. B.

Literature note. Potential external comparison sources: Wang et al. [5]; NIST ASD [1].

C Exact Parity, Role and Occupancy Comparators

C.1 Integer comparators

On each complete reference group C_{2m}, $m = 3, \ldots, 11$:

1. Exactly m seats have odd local parity and m even local parity. Against the comparator $\widehat{O}_m = m$,
$$\Delta^{\mathbb{Z}}_{\mathrm{parity},m} = \widehat{O}_m - O_{\mathrm{even},m} = 0.$$
Parity does not determine the assigned operational class.

2. The declared role projection has $m-1$ first-branch seats, one hinge seat, $m-1$ counterbranch seats and one boundary seat. Thus
$$\Delta^{\mathbb{Z}}_{\mathrm{branch},m} = (m-1) - (m-1) = 0.$$
These are counts of visual roles, not traversed edge events.

3. The terminal group has 14 labelled slots and capacity 22:
$$\Delta^{\mathbb{Z}}_{\mathsf{C}_{22}} = 14 - 22 = -8.$$
The difference counts eight unfilled reference slots and is not a retention or physical-cutoff residual.

C.2 Signed-magnitude record

The comparator representation is

$$\Delta_k^{\mathbb{Z}} = s_k^{(3)} m_k, \qquad s_k^{(3)} \in \{-1, 0, +1\}, \quad m_k = |\Delta_k^{\mathbb{Z}}|.$$

The two balanced comparisons give $\delta_3 = (0, 0)$; the terminal occupancy comparison gives $\delta_3 = (-1, 8)$. These values express the declared integer counts, not a native phase-cycle measurement.

Provenance. (11), (12), Main §5.7.13; released AOD Manual v41.02, App. C, for the signed comparator convention.

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[2] Reginald Poyau. Axiomatic fundamentalism (AF): A logical protocol for traceable research (all versions), 2025. All versions. DOI: `https://doi.org/10.5281/zenodo.17561186`.

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[5] M. Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi. The AME 2020 atomic mass evaluation (II). *Chinese Physics C*, 45:030003, 2021.